# Artificial Vision- Proposal

## Visualizing the Grid: Power Quality Disturbance Classification via Time-Frequency Spectrograms

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INTRODUCTION

# The Challenge of Modern Grid Complexity

Modern electrical grids face increasingly sophisticated power quality disturbances that traditional detection methods struggle to identify.

### Harmonics

Non-linear loads distort waveforms

### Voltage Sags

Brief drops damage equipment

### Swells

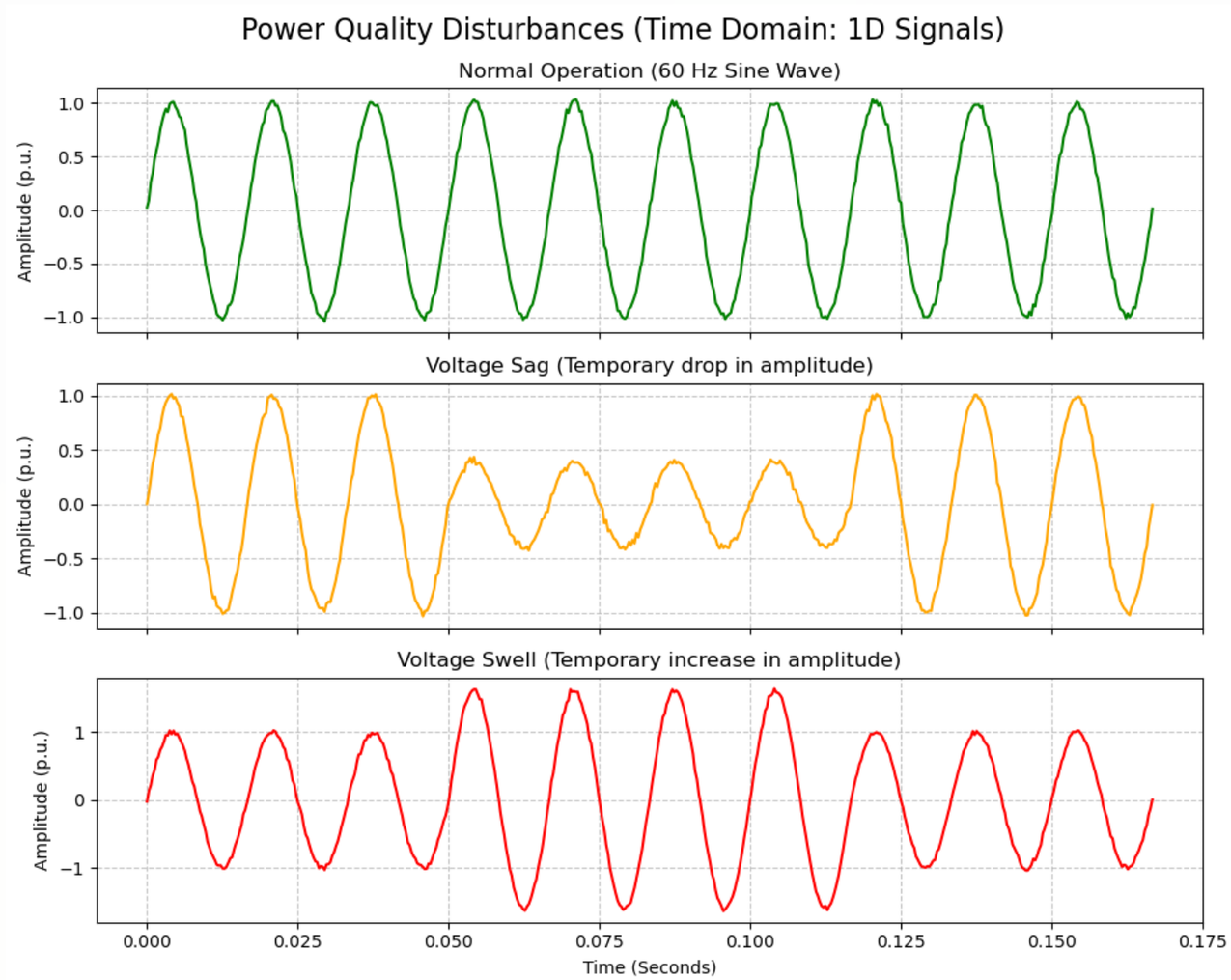
Transient overvoltages

# 1D dataset research

**SEED - Power Quality Disturbance Dataset**
URL:
https://www.kaggle.com/datasets/sumairaziz/seed-power-quality-disturbance-dataset

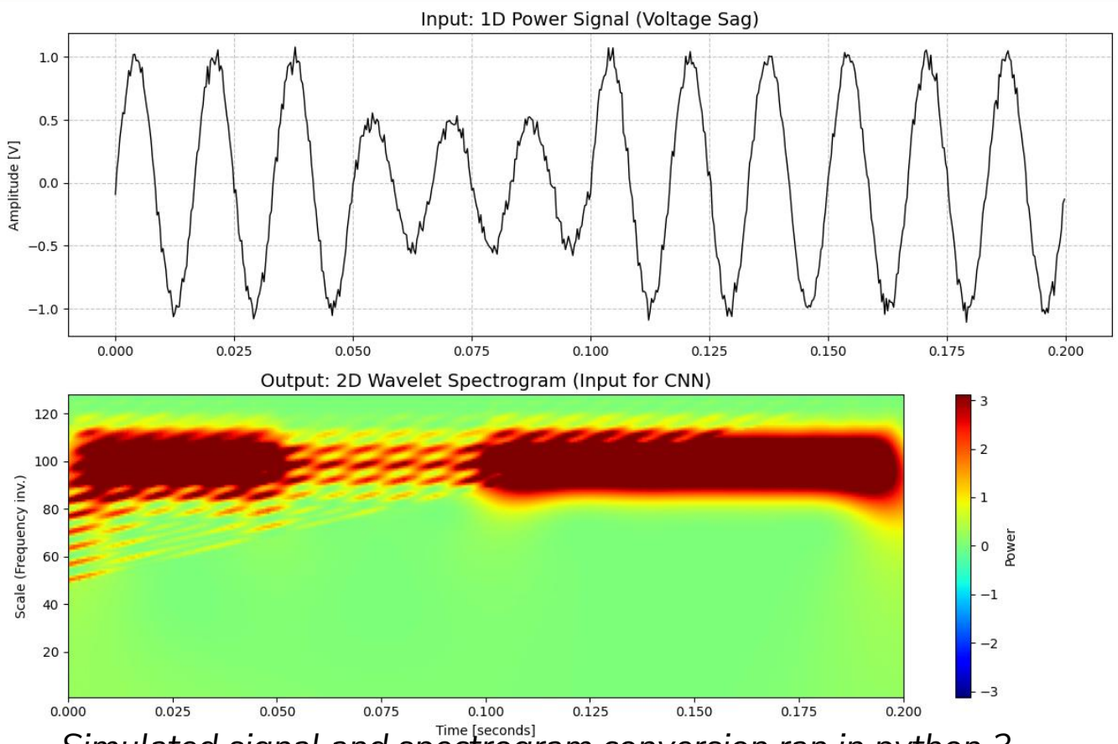
**Power Quality Classification Dataset - 1**
URL:
https://www.kaggle.com/datasets/jaideepreddykotla/powerqualitydistributiondataset1

**Generated Power Quality Disturbances**
Following IEEE standards



JUSTIFICATION

# The Visual Approach: Converting Signals to Images




*Simulated signal and spectrogram conversion ran in python 3.*

*ASSERTA FPGA as an example of an edge device, extracted from* [FPGA - INTESC]

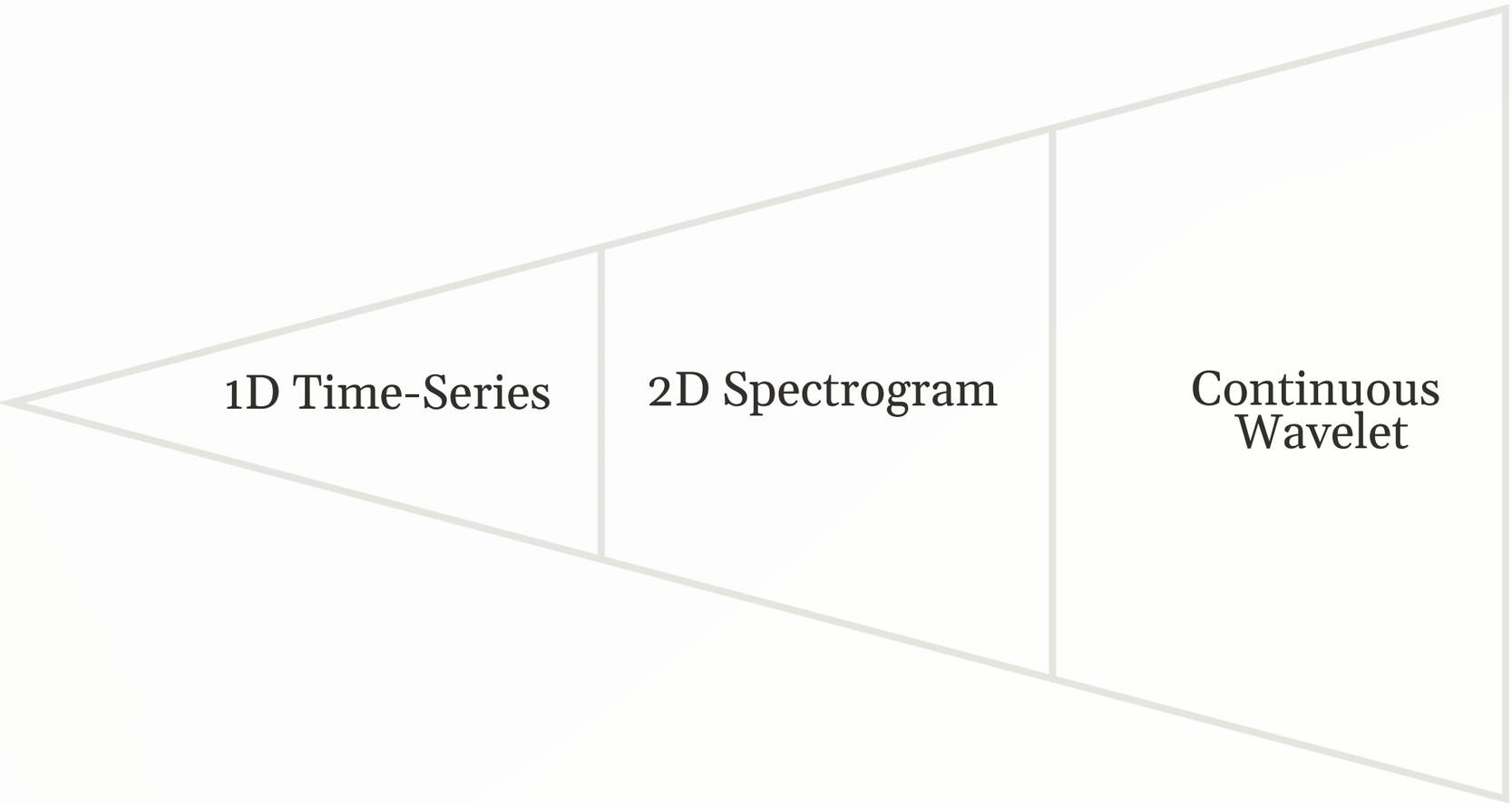
The optical systems and smart sensors in modern grids deliver complex electrical signals that require sophisticated analysis.

- **Why Images?** 2D representations allow us to visualize the "texture" of a fault (e.g., the specific visual signature of a harmonic).

- **Why Comparative Analysis?** Comparing a linear model (SVM) with a non-linear model (CNN) will validate whether the computational cost of Deep Learning yields a statistically significant improvement over simpler methods for this specific domain.

- **Application:** This visual approach is highly parallelizable and suitable for future deployment on embedded hardware (e.g., edge AI devices) for real-time monitoring.

# The Visual Approach: Converting Signals to Images




## Core Innovation

We transform time-series electrical signals into 2D spectrograms using techniques like **Continuous Wavelet Transform (CWT)**.

This reveals both temporal and frequency characteristics simultaneously, creating visual "fingerprints" of power quality disturbances.

STATE OF THE ART

# Finding the Sweet Spot

| Approach | Representative Reference | Limitation | The Proposed Solution |
|---|---|---|---|
| **1D Lightweight CNN** | Bai et al. (2026) | Fast but may miss complex time-frequency correlations. | **2D Spectrograms** capture rich "texture" information that 1D filters miss. |
| **Vision Transformers** | Anwar et al. (2025) | Computational overkill; requires massive data. | **Standard CNN** is more efficient for limited datasets while still being non-linear. |
| **Manual Features** | Guo et al. (2023) | Rigid; fails if noise patterns change. | **Deep Learning (CNN)** automates feature extraction, adapting to new noise patterns. |

# Research Hypothesis

> "Transforming one-dimensional electrical signals into two-dimensional spectrograms using Wavelet Transforms will allow a Convolutional Neural Network (CNN) to extract non-linear spatial features, thereby achieving higher classification accuracy for Power Quality disturbances compared to a linear Support Vector Machine (SVM) trained on the same spectral data."

OBJECTIVES

# Research Objectives

## General

To develop and validate a computer vision-based framework for classifying Power Quality disturbances by converting electrical time-series signals into spectrogram images.

## Specifics

1

### Dataset Curation

Compile and preprocess existing datasets like IEEE 1159 standard power quality disturbance signals

2

### Spectrogram Generation

Apply CWT to transform signals into time-frequency representations

3

### CNN Training

Develop and optimize convolutional neural network architecture

4

### Comparative Analysis

Benchmark CNN performance against SVM baseline classifier

| Phase | Activity | Description |
|---|---|---|
| **Phase 1** | Data Acquisition | Gather dataset containing labeled PQ events (Sags, Swells, Harmonics). Perform initial data cleaning and normalization. |
| **Phase 2** | Signal Processing | Apply Wavelet Transforms to the time-series data to generate 2D Spectrograms (Time-Frequency representation). Save these as image datasets. |
| **Phase 3** | Model A (Baseline) | Vectorize the spectrogram data and train a linear Support Vector Machine (SVM) classifier. |
| **Phase 4** | Model B (Proposed) | Design a CNN architecture. Train the model on the spectrogram images. |
| **Phase 5** | Validation & Analysis | Test both models on a hold-out test set. Generate confusion matrices and compare performance metrics. |
| **Phase 6** | Reporting | Document findings, visualize specific fault patterns "seen" by the CNN, and finalize the proposal report. |

# Expected Outcomes

## Explainability

Visual fault patterns enable human interpretation and trust

## Accuracy

Superior classification performance on complex disturbances

## Deployability

Practical solution for real-time grid monitoring systems

# References


[1] I. S. Samanta *et al.*, "A Comprehensive Review of Deep-Learning Applications to Power Quality Analysis," Jun. 01, 2023, *MDPI*. doi: 10.3390/en16114406.

[2] F. Guo, J. Li, X. Xu, Y. Zhu, X. Luo, and W. Qian, "Classification of Transient Power Quality Disturbances Based on Digital Image Processing Techniques," in *2023 8th International Conference on Power and Renewable Energy, ICPRE 2023*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 492–498. doi: 10.1109/ICPRE59655.2023.10353758.

[3] E. Perez-Anaya, A. Y. Jaen-Cuellar, D. A. Elvira-Ortiz, R. de J. Romero-Troncoso, and J. J. Saucedo-Dorantes, "Methodology for the Detection and Classification of Power Quality Disturbances Using CWT and CNN," *Energies (Basel).*, vol. 17, no. 4, Feb. 2024, doi: 10.3390/en17040852.

[4] A. A. Memon, M. A. Koondhar, S. F. Al-Gahtani, Z. M. S. Elbarbary, and Z. M. Alaas, "Comprehensive review of power quality disturbance detection and classification techniques," *Computers and Electrical Engineering*, vol. 126, Aug. 2025, doi: 10.1016/j.compeleceng.2025.110512.

[5] M. H. Anwar, M. M. A. Baig, A. J. Shaikh, and A. G. Abro, "Detection and classification of power quality disturbances: Vision transformers vs. CNN," *AIMS Energy*, vol. 13, no. 5, pp. 1052–1075, Jan. 2025, doi: 10.3934/energy.2025039.

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# THANKS!